

CHAPTER 2 - EVOLUTION OF MANAGEMENT

2.1. Pre classical Management concepts

2.1.1. Management in antiquity

Although the systematic study of management is relatively new, essentially the product of the beginning of the twenty-century, the concept and practice of management is not new. It had existed and been practiced since the beginning of man's cooperative action and may be traced back to the earliest records of history.

There was no systematic study of management in the periods before the 19 century. It was characterized by :

- Inadequacy of records or none at all.
- Poor communication, virtually with no exchange of ideas and practices.
- Business activities were limited and profit was very low.
- Absence of large business organizations.
- Failure to analyze the reasons for lack of success or low profit from earlier experiences, and
- Making managerial decisions on a trial and error basis.

However, there is various evidence about the existence of management practices in the antiquity.

1. SUMERIANS:

One of the earliest civilization know to man and employed the concept of management is that of the Sumerians. The famous contribution of Sumerians was the development of a written language in response to their need for managerial control was caps process around 3000BC. For centuries writings were a managerial tool for commerce, a matter of contract and bills of shipments and receipts, an instrument of religious records, an attempt to preserve magic formulas, ceremonial procedures, scared legends, and prayers. All these writing provide illustrations of managerial control since 3000BC.

2. THE EXODUS OF ISRAELITES (BIBLE EVIDENCE)

Additional evidence to illustrate the fact that interest in management as a process existed in antiquity can be found in the Bible. Exodus 18, verses 13 through 26, describes how Jethro, who

was Moses' father-in-law, provided Moses with several recommendations which are the early forerunners of our management concepts. For example, following some observations, Jethro advised Moses that what he was doing was entirely too much for one person and suggested several specific steps which should be taken to relieve Moses of this burden. One of his recommendations was that, as the Bible says, certain ordinances and laws should be taught to the people. Of course, if you look at this in light of modern management concepts, then you can see that what we are talking about, or what Jethro was talking about, is a counterpart of organizational statements of policies and procedures. Secondly, Jethro recommended to Moses that leaders be selected and assigned as they were to be "rulers of thousands," "rulers of hundreds," "rulers of fifties," and "rulers of tens." Here again, in modern managerial concepts, what Jethro was describing and recommending to Moses is now referred to as delegation of authority. Jethro's third recommendation to Moses was that these rulers should administer all routine matters and should bring to Moses only the important questions. The procedure of attending to the exception that does not conform to the expected results is known in present day managerial terms as the "principle of exception." But even while there is evidence of some management process in the old days, the real systematic study and development of formal literature and management did not appear until much later.

3. THE ANCIENT EGYPT:

Structures: the Egyptians built great pyramids from 500-525 BC. The pyramids were built with a technology comparable with modern standards. This indicates a mute testimony of the managerial and organizational abilities of ancient Egypt. Egyptians understood the importance of planning, organizing, controlling, division of labor, and the use of staff-advice as a valuable means for constructing magnificent structures. They also recognized and appreciated the values of spelling out job description and assignment of authority and responsibility.

Government: The study of the Egyptian government in the old, middle kingdom and in the new empire illustrates how they applied the principles of control. They exercised a highly decentralized government from 2160-1788 BC and a highly centralized government from 1530-1050 BC.

4. BABYLONIANS (200-1700 BC)

The most significant contribution of the Babylonians was king Hammurabi's code. Which contained 285 laws, to keep peace along the valleys of Tigris and Euphrates? These laws cover personal property, family, real estate, and trade. The Babylonians recognized some of the laws applicable to business and guide managerial decisions.

Look at the following:-

1. Developed control mechanisms: "If a merchant lends to an agent and the agent return to the merchant whatever the merchant had given to him. If the merchant deny receiving what the agent has given to him, that agent shall call the merchant to account in **the presence of God and witness** and the merchant shall give to him six fold the amount he obtained."
2. Introduced the concept of responsibility: "The mason who builds a house which falls dawn and kills the inmate shall be put to death". "If a man hires a tenant to oversee his farm and furnish his the seed grain and entrust him with oxen and contract with him to cultivate the field and that man steals either the seed or the crop and found in his possession, shall cut off his fingers."
3. Set minimum wage rate: "A man hiring a field laborer shall pay him 8 Gus of grain per year. Generally, Hammurabi not only provided management guidelines through his code but also a clear illustration of the importance of effective leadership style.

5. ROMANS

1. **Empire Structure:** Romans had been able to manage very large territory and people during their time. They built such giant government and military structure that were so successful and operated for so many years. They gained control over 50 million people extending from Great Britain to Syria including Europe and North Africa. They reinforced decentralized government by centralized army as the most effective way to organized and control their colonies.

The most effective management was put forth by Dioclerian's reorganization of the empire by abandoning the old organization. First, he divided the empire into 100 provinces. Then, each of these was grouped into one of 13 dioceses. In turn, the dioceses were grouped into four major geographical divisions, which are accountable to the emperor. Under this organization it was extremely difficult for anyone to defy imperial authority. To further

enhance his power, the emperor delegated only civil governmental authority to the governors and military forces were stationed under his control.

2. The Roman Catholic Church: This was the most effective formal organization in the western civilization and it has survived the test of time. The church has made significant contributions to management theories in the following areas.

- Hierarchy of authority: An organizational structure had been designed and scalar chain of command established from the people through the bishops to the priests and the laity.
- Specialization of activities along the functional lines and the development of the idea of job descriptions. In the second century two management historians have noted. “The bishop was the pastor, the presbyters constituted his council, and the deacons supervise the poor and the sick.” This indicates that there had been assignment of certain tasks to specific individuals in the church.
- The staff concept was developed from the advisory services performed by various individuals and committees. The concept of staff independence was also employed in the Catholic Church. This is because many subordinates feel that their boss does not listen to them.

6. CHINESE:

The Chinese also made contributions to management thought. One of the most famous is found in the ancient records of Mencius and Chow written around 1100-500 BC. It indicates that the Chinese were aware of certain principles bearing on planning, organizing, directing, and controlling. The constitution of Chow was a directory of all civil servants not clear to the emperor through a hierarchical relationship that extends from the prime minister down to the household servants with their specific jobs and responsibilities. The Chinese had also recognized the need for methodological selection, staffing, and specialization of employees that was carried out through their civil service system. Emperors back to 2350-2256 BC made constant use of advisory staff in every major decision.

7. GREEKS:

Greeks exhibit a real management skill and capacity in the operation of trading companies. They introduced the science and art of management. They recognized the means to maximize output through the use of uniform methods and motion study. They introduced standard motions, rhythm system of work and work tempos in harmony with music. Individuals, such as Xenophon recognized management as a separate art in 400 BC. Socrates showed the universality of management, the importance of employee selection, specialization of labor, and delegation of authority in 400BC. Plato also described specialization further in 350BC.

2.1.2 Early contributors

Many theorists and practitioners in the pre-classical period in the middle and late 1800s contributed valuable ideas that laid the basis for subsequent, broader inquiries into the nature of the management. Five principal pre-classical contributors of management thought in the early era: Robert Owen, Charles Babbage, Andrew Ure, Charles Dupin and Henry Robinson Towne. Nevertheless, four major contributors can be identified in this era *of management in the early factory system*: Robert Owen, Charles Babbage, Andrew Ure, and Charles Duoin. All were forerunners of latter developments in management.

Andrew Ure (1778-1857), French engineer and was one of the world's first professor to teach management principles, at Anderson's College in Glasgow, during the early seventeenth century. Ure argued that the factory system produces numerous advantages for society, far offsetting the problem it presents. He also believed that education was the answer in reducing the culture shock of industrialization.

Charles Dupin (1784-1873), Dupin was a French engineer and professor who pioneered industrial education in France. He was one of the first to advocate teaching broader management skills in college. He emphasized that managers needed more than technical skills to maximize output.

Sun Tzu; the oldest military treatise in the world is "the art of war", written by sun tzu, a chinese military writer, around 500B.C. this treatise illustrates that strategic planning, leadership, and effective management of people were all basic areas of interest to early military leaders.

Sun Tzu wrote on;

- Directing; if the words of command are not clear and distinct, if orders are not

thoroughly understood, the general is to blame. But if his orders are clear and the soldiers however disobey, it is the default of their officers.

- Planning; now the general who wins a battle makes many calculations in his temple where the battle is fought. The general who loses the battle make few calculations beforehand. It is by attention to this point that I can see who is likely to win or lose.

Nicco Machiavelli; born in Florence Italy, he was a diplomat, political philosopher, and civil servant of the Florence republic. Later Machiavelli lost his job and was exiled. During his exile, he wrote a book “the prince” which presents broad management principles to which he believed all leaders should subscribe.

Set four principles of leadership

- ❖ Mass content- every leader should realize that authority should emanates from the bottom and no one is a leader unless the followers agree.
- ❖ Strive for cohesiveness-a leader should ensure this in the organization by rewarding friends and supporters, and thus maintaining their allegiance.
- ❖ A will to survive- leaders must have a will to survive to keep them alert and ever prepared to protect their position should danger strike.
- ❖ Set an example-the leader has to set an example for the followers so that they can identify with him or her.

James Watt and Mathew Robinson Boulton; they contributed to the development of management thought by following certain management techniques in their engineering factory. They are;

- Production planning
- Standardization of components
- Provision of welfare for personnel
- Maintenance
- Scheme for executive development
- Marketing research and forecasting
- Elaborate statistical records

Adam Smith; the economic facet of management thought.

Adam Smith put the basic tenets of capitalism in his writing ‘an inquiry in to the nature and causes of the wealth of nations. The basic tenets of capitalism in Adam Smith’s work include;-

- Natural laws of demand and supply and the free market competition will efficiently regulate the flow of resource in a society.
- All individuals have the right to accumulate individual wealth.
- All individuals have the right to private ownership of property
- Division of labor will lead to greater gains in productivity

Division of labor, also called specialization, is the simple idea of breaking an entire job in to its component parts and assigning each specific task to an individual worker.

Smith described in detail the outcome and the advantage of division of labor in the first three chapters of *the wealth of nations*;

- Increased output because of increasing expertise by each worker as each concentrate on a smaller component of the task.
- Time saving
- The investment of specialized machines for each component of the overall work.

James Montgomery. He was a textile owner-manager in Scotland, actually wrote the first management texts for his and other companies. They were very technical and narrow in style, but addressed the problem of planning, organizing, and controlling in early factories.

2.1.3 Early works & Contributions of Charles Babbage:

One of the fore-runners of scientific management was Charles Babbage (1792-1871), Father of modern computing, an English mathematician, and focused his attention on efficiencies of production. He wrote a book entitled on *economy of machinery and manufacturer*, which reflects his view on management and manufacturing process, including in his belief that the division of labor is a necessary aspect of all developed economic systems. He also proposed observational procedures for studying manufacturing operations.

Charles Babbage is noted for having laid much of the ground work that later became known as scientific management; and he has regarded as the patron saint of operations research/management science.

Babbage’s invention includes

- The first meticulous calculator

- Rudimentary computer that included some aspects of contemporary computer hardware. One of the projects that Babbage worked on throughout his life, and unfortunately was never successful in being able to fully develop, was what he called the "difference engine." This invention is considered to be the forerunner of our modern data processing equipment. That actually performed mathematical calculations and statistical probabilities.
- Forerunner of time & motion study; he began to search for methods of improving work productivity with the use of work measurement and cost determination.
- Realizing the importance of the human factor; he believed that management and employees' interest are closely linked. As a result Babbage proposed profit sharing, bonus, and employee's suggestion plans-techniques still used today.
- Babbage's interest in production resulted in two contributions that remain as valid as today as when he presented them; the importance of dividing and assigning labor on the basis of skill, and means of determining the feasibility of replacing manual operations with machinery.

❖ **Division of labor;** Babbage presented a keen analysis of the alternative methods that are available for making pins. In this analysis of making pins by hand, he offered convincing data to show that a division of labor on the basis of skill is an economic necessity. Babbage pointed out that dividing labor by its level of skill;

- ✓ Reduces learning time since a given worker must learn only one operation rather than many operations.
- ✓ Saves manufacturing time, too, because less time is lost changing from one operation to another.
- ✓ Allow a high degree of skill to be acquired since each worker is learning only one task and repeating it many times.
- ✓ Opens up the possibility of developing specialized tools and equipment to further increase efficiency.
- ✓ Enable the manufacturer to acquire and pay for only the exact amount of skill required for each operation.

❖ **Manual versus automatic operations;** following his art of making pins by hand, Babbage mentioned briefly a new American process of making pins by

machines. In his analysis, he suggested several questions, the answers to which helped determine whether or not the introduction of the machine method was desirable. Those questions were:

1. To what defects pins by machine are liable?
2. What advantages do they possess over those made in the usual way?
3. What is the prime cost of a machine for making them?
4. What is the expense of keeping it in repair?
5. What is the expense of moving and attending to it?

If we take these same questions in light of today's machines and automation, we find that they are as important today as they were when they were originally asked 150 years ago. The importance lies in the fact that they raise questions concerning quality, the original cost of the machine, and the cost of operating the machine, including labour and maintenance. Much of management's efforts today are directed toward obtaining answers to these questions.

2.1.4 Times of Robert Owen & the management era

Robert Owen (1771-1858), a British Industrialist, reformer, and was one of the first manager to recognize the importance of an organization's human resource. Until his era, factory workers were generally viewed in the same way that machinery and equipment were. He was primarily interested in employee morale.

Important contributions of Owen

- 1. Introducing new ideas of human relations;** Owen initiated shorter working hour, housing facilities, training of workers in hygiene, education of their children, provision of canteen/meal services, etc in his early factory. He firmly believed that worker's performance in industry was influenced by working conditions and treatment of workers.
- 2. Proposing a legislative report that would limit the hour and usage of child labor;** Owen's proposal was viewed as a radical at that time. The proposed factory bill comprised of prohibiting employment of children under the age of ten, and limiting hours for all children to 10 ½ hours per day with no night work. The bill became law six years later, but was limited to cotton mills, reduced the age limit to nine, and included no provisions for inspections; therefore, the law had impact. Feeling frustrated in his attempts to reform Britain Owen traveled to America in 1824.

3. **The value of investment in human resource;** he claimed that returns from human recourse investment would far exceed those in machinery and equipments. He applied his belief on his own plants in Scotland and claimed that as a result of attentions to the needs of his personnel, he received over 50% return in his investment.
4. **Organizing cooperatives and trade union movement;** owen suggested the establishment of 'village cooperatives' based on common sharing of output of their inhabitants. On the other hand, he did not like the process of industrialization-especially with regard to specialization and division of labor. Owen viewed industrial workers as powerless before the onslaught of the factory system. He sounded much like the contemporary advocates of job enrichment as he talked of the industrial revolution destroying morale, sense of purpose, and social structure.

Now, Robert owen is considered as a forerunner of behavioral school because of his concern for human welfare.

2.2 The Era of Classical Management

2.2.1 Scientific Management Theory

Productivity emerged as a serious problem during first years of this century. Business was expanding and capital was readily available, but labor was in short supply. Therefore, managers begin to search for different ways to utilize the existing labour more efficiently. In response to this need, experts began to focus their research on ways to improve the performance of individual works and this leads to the development of scientific management. Scientific management is concerned with improving the performance of individual workers. Some of the earliest advocates of scientific management include:

- ✓ Fredrick Winslow Taylor (1856-1915)
- ✓ Henry L. Gantt (1861-1919)
- ✓ Frank Gilbreth (1868-1924) and Lillian Gilbreth (1878-1972)

These people have looked at determining the scientifically best methods for performing any task, for selecting, training and motivating workers. Among these Taylor has played dominant role.

Fredrick Winslow Taylor (March 20, 1856 - March 21, 1915) gave up going to college and started out as an apprentice pattern maker and machinist in 1875, joined the Midvale Steel Company in Philadelphia as a machinist in 1878, and rose to the position of chief engineer after

earning a degree in engineering through evening study. He invented high-speed steel-cutting tools and spent most of his life as a consulting engineer. Taylor is generally acknowledged as "Father of Scientific Management." Taylor's famous work entitled The "Principles of Scientific Management" was published in 1911.

Probably no other person has had a greater impact on the early development of management. His experience as an apprentice, a common labourer, a foreman, a master mechanic, and then the chief engineer of a steel company gave Taylor ample opportunity to know first-hand the problems and attitudes of workers and to see the great possibilities for improving the quality of management.

During his experience at Midvale Steel Company, he worked with production workers and keenly observed. However, it did not take much time for him to figure out that many workers were deliberately working at a pace slower than their capabilities. Then he called the tendency of restricting output as "Soldiering". Taylor studied and timed each element of the steelworkers' job. He determined what each worker should be producing, and then he designed the most efficient way of doing each part of the overall task. Next he implemented a piecework pay system. Rather than paying all employees same wage, he began increasing the pay of each worker who met and exceeded the target level of output set for his or her job.

After Taylor left Midvale, he worked as a consultant for several companies, including Simonds Rolling Machine Company and Bethlehem Steel. At Simonds he studied and redesigned jobs, introduced rest periods to reduce fatigue, and implemented a piecework pay system. The results were higher quality and quantity of output and improved morale. At Bethlehem Steel, Taylor studied efficient ways of loading and unloading rails and applied his conclusions with equally impressive results.

During these experiences, he discovered that production and pay were poor, inefficiency and waste were prevalent and most companies has tremendous unused potential. At last he concluded that the following are the major reasons for the inefficiency.

- ❖ Absence of standards of performance.
- ❖ Absence of motivating pay system.
- ❖ Absence of specialization.
- ❖ Absence of clearly distinguished responsibility of management and workers.
- ❖ Absence of harmonious relationship between management and subordinates.

In response to this, Taylor formulated the basic ideas that he called scientific management. As he conducted various experiments during this process which forms the basis of scientific management. It implies application of scientific principles for studying & identifying management problems.

According to Taylor, “Scientific Management is an art of knowing exactly what you want your men to do and seeing that they do it in the best and cheapest way”. In Taylor's view, if a work is analysed scientifically it will be possible to find one best way to do it. Hence scientific management is a thoughtful, organized, dual approach towards the job of management against hit or miss or Rule of Thumb. According to Drucker, “The cost of scientific management is the organized study of work, the analysis of work into simplest element & systematic management of worker's performance of each element”.

2.2.2 Classical Organizational Theories

Whereas scientific management theory is concerned with increasing the performance of individual workers but Classical organizational theory focuses on managing the total organization. Classical organization theory is also called administrative theory. The administrative branch emphasized on efficiency and productivity in running factories and business.

French man believed that management was not a person talent that somebody had by birth and other did not. Management required specific skills that could be learned, taught and gained by personal experience.

Henry Fayol, a French industrialist concentrated on the administration aspect of scientific management. His contributions are concentrated in his famous book “The General and Industrial Administration”. Fayol's famous book falls into two parts. The first is concerned with the theory of administration in which Fayol divided the total industrial activities into six categories which are given below:

1. Technical (Production, Manufacture).
2. Commercial (Buying, Selling and Exchange).
3. Financial (Search for and optimum use of capital).
4. Security (Protection of property and persons).
5. Accounting (Balance sheets, Cost and Statistics).
6. Management (Planning, Organizing, Coordinating, Directing and Controlling).

The second is concerned with the fourteen principles of management.

A principle refers to a fundamental truth. It establishes cause and effect relationship between two or more variables under given situation. They serve as a guide to thought & actions. Therefore, management principles are the statements of fundamental truth based on logic which provides guidelines for managerial decision making and actions. These principles are derived: -
On the basis of observation and analysis i.e. practical experience of managers. By conducting experimental studies.

There are 14 Principles of Management described by Henri Fayol.

1. Division of Labor

Henry Fayol has stressed on the specialization of jobs. He recommended that work of all kinds must be divided & subdivided and allotted to various persons according to their expertise in a particular area. Subdivision of work makes it simpler and results in efficiency. It also helps the individual in acquiring speed, accuracy in his performance. Specialization leads to efficiency & economy in spheres of business.

2. Party of Authority & Responsibility

Authority & responsibility are co-existing. If authority is given to a person, he should also be made responsible. In a same way, if anyone is made responsible for any job, he should also have concerned authority. Authority refers to the right of superiors to get exactness from their subordinates whereas responsibility means obligation for the performance of the job assigned. There should be a balance between the two i.e. they must go hand in hand. Authority without responsibility leads to irresponsible behavior whereas responsibility without authority makes the person ineffective.

3. Principle of One Boss

A sub-ordinate should receive orders and be accountable to one and only one boss at a time. In other words, a sub-ordinate should not receive instructions from more than one person because -

- It undermines authority
- Weakens discipline
- Divides loyalty
- Creates confusion
- Delays and chaos
- Escaping responsibilities

- Duplication of work
- Overlapping of efforts

Therefore, dual sub-ordination should be avoided unless and until it is absolutely essential. Unity of command provides the enterprise a disciplined, stable & orderly existence. It creates harmonious relationship between superiors and sub-ordinates.

4. Unity of Direction

Fayol advocates one head one plan which means that there should be one plan for a group of activities having similar objectives. Related activities should be grouped together. There should be one plan of action for them and they should be under the charge of a particular manager.

According to this principle, efforts of all the members of the organization should be directed towards common goal. Without unity of direction, unity of action cannot be achieved.

In fact, unity of command is not possible without unity of direction.

Therefore it is obvious that they are different from each other but they are dependent on each other i.e. unity of direction is a pre-requisite for unity of command. But it does not automatically comes from the unity of direction.

5. Equity

Equity means combination of fairness, kindness & justice. The employees should be treated with kindness & equity if devotion is expected of them. It implies that managers should be fair and impartial while dealing with the subordinates. They should give similar treatment to people of similar position. They should not discriminate with respect to age, caste, sex, religion, relation etc. Equity is essential to create and maintain cordial relations between the managers and sub-ordinate. But equity does not mean total absence of harshness. Fayol was of opinion that, “at times force and harshness might become necessary for the sake of equity”.

6. Order

This principle is concerned with proper & systematic arrangement of things and people. Arrangement of things is called material order and placement of people is called social order.

Material order- There should be safe, appropriate and specific place for every article and every place to be effectively used for specific activity and commodity.

Social order- Selection and appointment of most suitable person on the suitable job. There should be a specific place for everyone and everyone should have a specific place so that they can easily be contacted whenever need arises.

7. Discipline

According to Fayol, “Discipline means sincerity, obedience, respect of authority & observance of rules and regulations of the enterprise”. This principle applies that subordinate should respect their superiors and obey their order. It is an important requisite for smooth running of the enterprise. Discipline is not only required on path of subordinates but also on the part of management. Discipline can be enforced if -

- There are good superiors at all levels.
- There are clear & fair agreements with workers.
- Sanctions (punishments) are judiciously applied.

8. Initiative

Workers should be encouraged to take initiative in the work assigned to them. It means eagerness to initiate actions without being asked to do so. Fayol advised that management should provide opportunity to its employees to suggest ideas, experiences & new method of work. It helps in developing an atmosphere of trust and understanding. People then enjoy working in the organization because it adds to their zeal and energy. To suggest improvement in formulation & implementation of place. They can be encouraged with the help of monetary & non-monetary incentives.

9. Fair Remuneration

The quantum and method of remuneration to be paid to the workers should be fair, reasonable, satisfactory & rewarding of the efforts.

As far as possible it should accord satisfaction to both employer and the employees.

Wages should be determined on the basis of cost of living, work assigned, financial position of the business, wage rate prevailing etc.

Logical & appropriate wage rates and methods of their payment reduce tension & differences between workers & management creates harmonious relationship and pleasing atmosphere of work.

Fayol also recommended provision of other benefits such as free education, medical & residential facilities to workers.

10. Stability of Tenure

Fayol emphasized that employees should not be moved frequently from one job position to another i.e. the period of service in a job should be fixed.

Therefore employees should be appointed after keeping in view principles of recruitment & selection but once they are appointed their services should be served.

According to Fayol. “Time is required for an employee to get used to a new work & succeed to doing it well but if he is removed before that he will not be able to render worthwhile services”.

As a result, the time, effort and money spent on training the worker will go waste.

Stability of job creates team spirit and a sense of belongingness among workers which ultimately increase the quality as well as quantity of work.

11. Scalar Chain

Fayol defines scalar chain as 'The chain of superiors ranging from the ultimate authority to the lowest’.

Every orders, instructions, messages, requests, explanation etc. has to pass through Scalar chain. But, for the sake of convenience & urgency, this path can be cut short and this short cut is known as Gang Plank.

A Gang Plank is a temporary arrangement between two different points to facilitate quick & easy communication as explained below:

Gang Plank clarifies that management principles are not rigid rather they are very flexible. They can be moulded and modified as per the requirements of situations

12. Sub-Ordination of Individual Interest to General Interest

An organization is much bigger than the individual it constitutes therefore interest of the undertaking should prevail in all circumstances.

As far as possible, reconciliation should be achieved between individual and group interests.

But in case of conflict, individual must sacrifice for bigger interests.

In order to achieve this attitude, it is essential that -

- Employees should be honest & sincere.
- Proper & regular supervision of work.
- Reconciliation of mutual differences and clashes by mutual agreement. For example, for change of location of plant, for change of profit sharing ratio, etc.

13. Spirit De' Corps (can be achieved through unity of command)

It refers to team spirit i.e. harmony in the work groups and mutual understanding among the members.

Spirit De' Corps inspires workers to work harder.

Fayol cautioned the managers against dividing the employees into competing groups because it might damage the moral of the workers and interest of the undertaking in the long run.

To inculcate *Espirit De' Corps* following steps should be undertaken -

- ✓ There should be proper co-ordination of work at all levels
- ✓ Subordinates should be encouraged to develop informal relations among themselves.
- ✓ Efforts should be made to create enthusiasm and keenness among subordinates so that they can work to the maximum ability.
- ✓ Efficient employees should be rewarded and those who are not up to the mark should be given a chance to improve their performance.
- ✓ Subordinates should be made conscious of that whatever they are doing is of great importance to the business & society.

He also cautioned against the more use of Britain communication to the subordinates i.e. face to face communication should be developed. The managers should infuse team spirit & belongingness. There should be no place for misunderstanding. People then enjoy working in the organization & offer their best towards the organization.

14. Centralization & De-Centralization

Centralization means concentration of authority at the top level. In other words, centralization is a situation in which top management retains most of the decision making authority.

Decentralization means disposal of decision making authority to all the levels of the organization. In other words, sharing authority downwards is decentralization.

According to Fayol, "Degree of centralization or decentralization depends on no. of factors like size of business, experience of superiors, dependability & ability of subordinates etc.

Anything which increases the role of subordinate is decentralization & anything which decreases it is centralization.

Fayol suggested that absolute centralization or decentralization is not feasible. An organization should strike to achieve a lot between the two.

2.2.3 Bureaucratic Management theory

Whereas scientific management focused on the productivity of individuals, the classical administrative approach concentrates on the total organization. The emphasis is on the development of managerial principles rather than work methods.

Contributors to this school of thought include Max Weber, Henri Fayol, Mary Parker Follett, and Chester I. Barnard. These theorists studied the flow of information within an organization and emphasized the importance of understanding how an organization operated.

In the late 1800s, Max Weber disliked that many European organizations were managed on a “personal” family-like basis and that employees were loyal to individual supervisors rather than to the organization. He believed that organizations should be managed impersonally and that a formal organizational structure, where specific rules were followed, was important. In other words, he didn't think that authority should be based on a person's personality. He thought authority should be something that was part of a person's job and passed from individual to individual as one person left and another took over. This non personal, objective form of organization was called a bureaucracy.

Weber believed that all bureaucracies have the following characteristics:

- A well-defined hierarchy. All positions within a bureaucracy are structured in a way that permits the higher positions to supervise and control the lower positions. This clear chain of command facilitates control and order throughout the organization.
- Division of labor and specialization. All responsibilities in an organization are specialized so that each employee has the necessary expertise to do a particular task.
- Rules and regulations. Standard operating procedures govern all organizational activities to provide certainty and facilitate coordination.
- Impersonal relationships between managers and employees. Managers should maintain an impersonal relationship with employees so that favoritism and personal prejudice do not influence decisions.
- Competence. Competence, not “who you know,” should be the basis for all decisions made in hiring, job assignments, and promotions in order to foster ability and merit as the primary characteristics of a bureaucratic organization.
- Records. A bureaucracy needs to maintain complete files regarding all its activities.

2.2.4 Total Appraisal of the Classical theories

I. Highlights the need for a scientific approach to management.

II. Points out that work methods often can be improved through study.

III. Identifies a number of important principles that are useful in running organization efficiently.

2.3 The Human Relations Movement

2.3.1. Behavioral school

It is also called Human Relations, Leadership or Behavioral Science Approach. It bears the existing and newly developed theories and methods of the relevant social sciences upon the study of human behavior ranging from personality dynamics of individuals to the relations of culture. Behaviourists' believe that those involved in the organization are the prime determinants of organizational and managerial effectiveness. Member of this school see human behavior as complex subject & as most vital aspect of management.

Features

- ✓ As management is the process of getting things done by people, managers should understand human behavior.
- ✓ Emphasis is put on increasing productivity through motivation and good human relations.
- ✓ Motivation, leadership, communication, participative management and group dynamics are the central core of this approach.

Uses

It suggests how the knowledge of human behavior can be used in making people more effective in the organization. An individual's behavior is not determined by organization factors alone but also by his attitude, pressure, conflicts of cultural environment etc. Hence these factors must be taken into account.

2.3.2. Elton Mayo and the Behavioral Schools

George Elton Mayo (26 December 1880 - 7 September 1949) was an Australian psychologist, sociologist and organization theorist.

He lectured at the University of Queensland from 1911 to 1923 before moving to the University of Pennsylvania, but spent most of his career at Harvard Business School (1926 - 1947), where he was professor of industrial research. On 18 April 1913 he married Dorothea McConnell in Brisbane, Australia. They had two daughters, Patricia and Gael.

Mayo is known as the founder of the Human Relations Movement, and is known for his research including the Hawthorne Studies and his book *The Human Problems of an Industrialized*

Civilization (1933). The research he conducted under the Hawthorne Studies of the 1930s showed the importance of groups in affecting the behaviour of individuals at work. Mayo's employees, Roethlisberger and Dickson, conducted the practical experiments. This enabled him to make certain deductions about how managers should behave. He carried out a number of investigations to look at ways of improving productivity, for example changing lighting conditions in the workplace. What he found however was that work satisfaction depended to a large extent on the informal social pattern of the work group. Where norms of cooperation and higher output were established because of a feeling of importance, physical conditions or financial incentives had little motivational value. People will form work groups and this can be used by management to benefit the organization.

He concluded that people's work performance is dependent on both social issues and job content. He suggested a tension between workers' 'logic of sentiment' and managers' 'logic of cost and efficiency' which could lead to conflict within organizations.

Disagreement regarding his employees' procedure while conducting the studies:

- The members of the groups whose behavior has been studied were allowed to choose themselves.
- Two women have been replaced since they were chatting during their work. They were later identified as members of a leftist movement.
- One Italian member was working above average since she had to care for her family alone. Thus she affected the group's performance in an above average way.

Summary of Mayo's Beliefs:

- Individual workers cannot be treated in isolation, but must be seen as members of a group.
- Monetary incentives and good working conditions are less important to the individual than the need to belong to a group.
- Informal or unofficial groups formed at work have a strong influence on the behaviour of those workers in a group.
- Managers must be aware of these 'social needs' and cater for them to ensure that employees collaborate with the official organization rather than work against it.
- Mayo's simple instructions to industrial interviewers set a template and remain influential to this day i.e. A. The simple rules of interviewing:-

1. Give your full attention to the person interviewed, and make it evident that you are doing so.
2. Listen - don't talk.
3. Never argue; never give advice.
4. Listen to: what he wants to say; what he does not want to say; what he cannot say without help.
5. As you listen, plot out tentatively and for subsequent correction the pattern that is being set before you. To test, summarize what has been said and present for comment. Always do this with caution - that is, clarify but don't add or twist. [SOURCE: papers held by Mayo's grand- daughter.]

2.3.3. Pros and Cons of the Behavioral Schools

Pros:

Observers are always a factor in their research, no matter how neutral their position. This influence on the group must be acknowledged during an analysis or implementation activity. The 'Hawthorne Effect' led to the development of 'Action Research', a methodology that assumed that understanding of situations and changes occur simultaneously.

Informal groups exert strong social controls over the work habits and attitudes of the individual employee. The strengthening of informal relations among organisation members, the development of 'hero' teams as well as the fostering of cultural aspects are important design factors that increase employees' feelings of belonging. However, informal groups can also be detrimental, particularly, when group norms prevent new ways of working.

- ✓ A business organization is basically a social system. It is not just a techno-economic system. Productivity gains are achieved not only by a more efficient layout of production line, but also by employee's mental states.
- ✓ The employee can be motivated by psychological & social wants. Because his/her behaviour is influenced by feelings, emotions & attitudes, thus economic incentives are not the only method to motivate employee.
- ✓ Management must learn to develop co-operative attitudes & not rely merely on command.

- ✓ Participation becomes an important instrument in human relations movement. In order to achieve participation, effective two-way communication network is essential.
- ✓ Productivity is linked with employee satisfaction in any business organization. Therefore management must take greater interest in employee interest.
- ✓ Group psychology plays an important role in any business organization. We must therefore rely more on informational group effort.
- ✓ The key to higher productivity lies in employee morale, high morale results in higher output.

Cons:

Mayo's work focused on the social end of the continuum to offset the dominance of Scientific Management theory. Since Mayo did not try to integrate his conclusions with Scientific Management, his recommendations remained uni-dimensional.

Mayo's interpretation that employees derive greater motivation from informal groups does not rule out other causes such as learning, expertise and reflection, all of which remained unexamined premises independent of Mayo's investigated interventions.

It is assumed that all organizational problems are amenable to solutions through human relations. This assumption does not hold well in practice. The human relationist saw only the human variables as critical & ignored other variables.

The human relationists are over emphasis on the group & group decision making. But in practice, groups may create over problems & collective decision making may not be possible.

2.3.4. The Hawthorn Study

Harvard Business School and the Hawthorne Experiments (1924-1933)

In the 1920s Elton Mayo, a professor of Industrial Management at Harvard Business School, and his protégé Fritz J. Roethlisberger led a landmark study of worker behaviour at Western Electric, the manufacturing firm of AT&T. Unprecedented in scale and scope, the nine-year study took place at the massive Hawthorne Works plant outside of Chicago and generated a mountain of documents, from hourly performance charts to interviews with thousands of employees. Harvard Business School's role in the experiments represented a milestone in the dawn of the human relations movement and a shift in the study of management from a scientific to a multi-disciplinary approach. Baker Library's exhaustive archival record of the experiments reveals the

art and science of this seminal behavioural study—and the questions and theories it generated about the relationship of productivity to the needs and motivations of the industrial worker.

I. Illumination Study, 1926

Research on productivity at massive manufacturing complexes like the Hawthorne Works was made possible through partnerships among industries, universities, and government. In the 1920s, with support from the National Research Council, the Rockefeller Foundation, and eventually Harvard Business School, Western Electric undertook a series of behavioural experiments. The first, a sequence of illumination tests from 1924 to 1927, set out to determine the effects of lighting on worker efficiency in three separate manufacturing departments. Accounts of the study revealed no significant correlation between productivity and light levels. The results prompted researchers to investigate other factors affecting worker output.

II. Relay Assembly Test Room Experiments

The next experiments beginning in 1927 focused on the relay assembly department, where the electromagnetic switches that made telephone connections possible were produced. The manufacture of relays required the repetitive assembly of pins, springs, armatures, insulators, coils, and screws. Western Electric produced over 7 million relays annually. As the speed of individual workers determined overall production levels, the effects of factors like rest periods and work hours in this department were of particular interest to the company.

In one of the studies, experimenters chose two women as test subjects and asked them to choose four other workers to join the test group. Together the women worked in a separate room over the course of five years (1927–1932) assembling telephone relays.

Output was measured mechanically by counting how many finished relays each dropped down a chute. This measuring began in secret two weeks before moving the women to an experiment room and continued throughout the study. In the experiment room, they had a supervisor who discussed changes with them and at times used their suggestions. Then the researchers spent five years measuring how different variables impacted the group's and individuals' productivity. Some of the variables were:

- Giving two 5-minute breaks (after a discussion with them on the best length of time), and then changing to two 10-minute breaks (not their preference). Productivity increased, but when they received six 5-minute rests, they disliked it and reduced output.
- providing food during the breaks

- Shortening the day by 30 minutes (output went up); shortening it more (output per hour went up, but overall output decreased); returning to the first condition (where output peaked).

Changing a variable usually increased productivity, even if the variable was just a change back to the original condition. However it is said that this is the natural process of the human being to adapt to the environment without knowing the objective of the experiment occurring. Researchers concluded that the workers worked harder because they thought that they were being monitored individually.

Researchers hypothesised that choosing one's own co-workers, working as a group, being treated as special (as evidenced by working in a separate room), and having a sympathetic supervisor were the real reasons for the productivity increase. One interpretation, mainly due to Elton Mayo, was that "the six individuals became a team and the team gave itself wholeheartedly and spontaneously to cooperation in the experiment." (There was a second relay assembly test room study whose results were not as significant as the first experiment.)

III. Bank wiring room experiments

This experiment was conducted with a view to systematically analyze the functioning of small group & its impact on individual behaviour as well as the causes which restrict output. The surprising result was that productivity actually decreased. Workers apparently had become suspicious that their productivity may have been boosted to justify firing some of the workers later on. The study was conducted by Mayo and W. Lloyd Warner between 1931 and 1932 on a group of fourteen men who put together telephone switching equipment. The researchers found that although the workers were paid according to individual productivity, productivity decreased because the men were afraid that the company would lower the base rate. Detailed observation between the men revealed the existence of informal groups or "cliques" within the formal groups. These cliques developed informal rules of behaviour as well as mechanisms to enforce them. The cliques served to control group members and to manage bosses; when bosses asked questions, clique members gave the same responses, even if they were untrue. These results show that workers were more responsive to the social force of their peer groups than to the control and incentives of management.

IV. The Interview Process

I think interviewing is a good idea. It helps some people get a lot of things off their chest.

Western Electric employee, in Comments and Reactions on Interviewing Program, ca. 1930

*Factory Cabling Department, ca. 1925*Assisting Mayo was his research assistant, Fritz Roethlisberger. Unassuming, bookish, and disciplined, Roethlisberger had studied philosophy at Harvard. He worked as a psychological counsellor for Harvard students and became known as an expert listener. Roethlisberger, who found himself “spellbound by Mayo’s...creative imagination and clinical insights,” would himself become one of Harvard Business School’s beloved and highly sought after professors.

Under Mayo and Roethlisberger’s direction, the Hawthorne experiments began to incorporate extensive interviewing. The researchers hoped to glean details (such as home life or relationship with a spouse or parent) that might play a role in employees’ attitudes towards work and interactions with supervisors. From 1928 to 1930 Mayo and Roethlisberger oversaw the process of conducting more than 21,000 interviews and worked closely training researchers in interviewing practices.

Mayo and Roethlisberger’s methodology shifted when they discovered that, rather than answering directed questions, employees expressed themselves more candidly if encouraged to speak openly in what was known as non-directed interviewing. “It became clear that if a channel for free expression were to be provided, the interview must be a listening rather than a questioning process,” a research study report noted. “The interview is now defined as a conversation in which the employee is encouraged to express himself freely upon any topic of his own choosing.

Interviews, which averaged around 30 minutes, grew to 90 minutes or even two hours in length in a process meant to provide an emotional release. The resulting records, hundreds and hundreds of pages in which employees disclose personal details of their day to day lives, offer an astonishingly intimate portrait of the American industrial worker in the years leading to and following the Depression. In a pre-computer age, thousands of comments were sorted into employees’ attitudes about general working conditions, specific jobs, or supervisors and among these categories into favourable and unfavourable comments used to support interpretations of the data. Both workers’ and supervisors’ comments would aid in the development of personnel

policies and supervisory training, including the subsequent implementation of a routine counselling program for employees.

In his autobiography *The Elusive Phenomena*, Roethlisberger wrote of grappling with objective, hard data versus subjective, soft data. “I felt very strongly,” he noted, “that in the gooey soft data there existed uniformities about human behaviour that had to be coaxed out by...the method of clinical observation and interviewing which I was advocating for the administrator to use. Roethlisberger discovered that what employees found most deeply rewarding were close associations with one another, “informal relationships of interconnectedness,” as he called them. “Whenever and where it was possible,” he wrote, “[employees] generated them like crazy. In many cases they found them so satisfying that they often did all sorts of non-logical things...in order to belong. In Mayo’s broad view, the industrial revolution had shattered strong ties to the workplace and community experienced by workers in the skilled trades of the 19th century. The social cohesion holding democracy together, he wrote, was predicated on these collective relationships, and employees’ belief in a sense of common purpose and value of their work.

2.4 The Modern Era

2.4.1 Characteristics of Modern Approaches to Management

Modern management approaches recognize that people are complex and variable. Employee needs change over time; people possess a range of talents and capabilities that can be developed. Organizations and managers, therefore, should respond to individuals with a wide variety of managerial strategies and job opportunities.

Some of the notable characteristics of the modern approaches to the organization are:

- ❖ A systems viewpoint
- ❖ A dynamic process of interaction
- ❖ Multileveled and multidimensional
- ❖ Multi-motivated
- ❖ Probabilistic
- ❖ Multi-disciplinary
- ❖ Descriptive
- ❖ Multivariable, and
- ❖ Adaptive

2.4.2 Systems Approach

Rather than dealing separately with the various segments of an organization, the systems approach to management views the organization as a unified, purposeful system composed of interrelated parts. This approach gives managers a way of looking at the organization as a whole and as a part of the larger, external environment. Systems theory tells us that the activity of any segment of an organization affects, in varying degrees, the activity of every other segment.

Production managers in a manufacturing plant, for example, prefer long uninterrupted production runs of standardized products in order to maintain maximum efficiency and low costs. Marketing managers, on the other hand, who want to offer customers quick delivery of a wide range of products, would like a flexible manufacturing schedule that can fill special orders on short notice. Systems oriented production managers make scheduling decisions only after they have identified the impact of these decisions on other departments and on the entire organization. The point of the systems approach is that managers cannot function wholly within the confines of the traditional organization chart. They must mesh their department with the whole enterprise. To do that, they have to communicate not only with other employees and departments, but frequently with representatives of other organizations as well. Clearly, systems managers grasp the importance of webs of business relationships to their efforts.

The **systems management theory** has had a significant effect on management science. A system is an interrelated set of elements functioning as a whole.

An organization as a system is composed of four elements:

- a. Inputs:** the various human, materials, financial, equipment, and informational resources required to produce goods and services.
- b. Transformation processes:** the organization's managerial and technological abilities that are applied to convert inputs into outputs.
- c. Outputs:** the products, services, and other outcomes produced by the organization.
- d. Feedback:** information about results and organizational status relative to its environment

2.4.3 The quantitative approach

During World War II, mathematicians, physicists, and other scientists joined together to solve military problems. The quantitative school of management is a result of the research conducted during World War II. The **quantitative approach** to management involves the use of quantitative techniques, such as statistics, information models, and computer simulations, to improve decision making.

2.4.4 The Contingency Approach

The contingency school of management can be summarized as an “it all depends” approach. The appropriate management actions and approaches depend on the situation. Managers with a contingency view use a flexible approach, draw on a variety of theories and experiences, and evaluate many options as they solve problems.

Contingency management recognizes that there is no one best way to manage. In the contingency perspective, managers are faced with the task of determining which managerial approach is likely to be most effective in a given situation. For example, the approach used to manage a group of teenagers working in a fast-food restaurant would be very different from the approach used to manage a medical research team trying to find a cure for a disease.

Contingency thinking avoids the classical “one best way” arguments and recognizes the need to understand situational differences and respond appropriately to them. It does not apply certain management principles to any situation. Contingency theory is recognition of the extreme importance of individual manager performance in any given situation. The contingency approach is highly dependent on the experience and judgment of the manager in a given organizational environment.

According to Donnelly, Gibson, and Ivancevich (1995), Managers around the globe use more of a contingency approach to survive. Sole reliance on classical or a behavioural or a management science approach is not sufficient for organizations. Thus, the contingency view has become relevant and prominent because of the following factors:

1. Increased globalization of enterprise and the need for more government-business

Alliances to compete internationally;

2. Demands for ethical and socially responsive leadership;
3. Changing demographics and skill requirements of the workforce;
4. The emergence of new organizational structures that emphasize speed in reacting to environmental changes; and
5. Changing needs, preferences, and desires of employees for job security, participation, ownership, and personal fulfilment (Donnelly et al., 1995, p. 21).

To sum up, the organizational environment theory (both the systems approach and the contingency approach) can provide valuable insights and complement the classical, behavioral, and management science approaches. The systems approach forces managers to recognize that organizations are systems made up of interdependent parts and that a change in one part affects other parts. It seeks to identify the characteristics of jobs, people, and organizations, allowing managers to see the interdependence between the various segments of an organization. The basic idea of the contingency approach is that there is no best way to plan, organize, lead, or control. Rather, managers must find different ways to fit different situations. As the contingency approach suggests, the student of management preparing to manage in the 21st century must learn multiple ways to compete, innovate, create, motivate, and lead.

2.4.5 The Modern Approaches

Modern management approaches respect the classical and human resource approaches to management. However, successful managers recognize that although each theoretical school has limitations in its applications, each approach also offers valuable insights that can broaden a manager's options in solving problems and achieving organizational goals. Successful managers work to extend these approaches to meet the demands of a dynamic environment.

Modern theories tend to be based on the concept that the organization is a system which has to adapt to changes in its environment. In modern theory, an organization is defined as a designed and structured process in which individuals interact for objectives (Hicks and Gullet, 1975). The

contemporary approach to the organization is multidisciplinary, as many scientists from different fields have contributed to its development, emphasizing the dynamic nature of communication and importance of integration of individual and organizational interests. These were subsequently re-emphasized by Bernard (1938) who gave the first modern and comprehensive view of management. Subsequently, conclusions on systems control gave insight into application of cybernetics. The operation research approach was suggested in 1940. It utilized the contributions of several disciplines in problem solving. Von Bertalanffy (1951) made a significant contribution by suggesting a component of general systems theory which is accepted as a basic premise of modern theory.